

## ATO Detection via Device Embeddings

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Can login device sequences catch account takeover in real time?

**Answer:** Yes — with the right signal and training configuration.

FastText on structured feature tokens achieves **AUC 0.985** on novel attacks at sub-millisecond inference latency.

Three experiments · Adversarial critique & debate · Peer review · Production analysis

[github.com/chris-santiago/ato-device-embeddings](https://github.com/chris-santiago/ato-device-embeddings)

# Which Signal Works?

Three experiments compared embedding strategies across attack types.

| Signal                        | Novel AUC         | Fleet AUC | Spoof AUC | Verdict                           |
|-------------------------------|-------------------|-----------|-----------|-----------------------------------|
| Feature tokens, mean-pool     | 0.999             | 0.994     | 0.869     | ✔ Recommended                     |
| Feature tokens, concat string | 0.997             | 0.998     | 0.782     | ⚠ Beats trivial on spoof (+0.032) |
| Word2Vec on device IDs        | —                 | 0.891     | —         | ✔ Offline fleet detection only    |
| FastText on device IDs        | silhouette -0.051 | —         | —         | ✘ Destroys cluster structure      |
| Trivial set-membership        | 0.750             | 0.750     | 0.750     | Fallback only                     |

**Key insight:** Mean-pool (+0.119 over trivial) decisively outperforms concat (+0.032) on spoof attacks — the hardest case, where the attacker matches the victim's OS, browser, and language and differs only on timezone, network, and screen.

# Why Mean-Pool Beats Concatenated Strings

Character n-grams in a concat string span feature boundaries ( `ari_ut` , `i_utc` ), injecting spurious signal that dilutes any single mismatched feature. Mean-pooling embeds each feature independently — a differing timezone contributes its full weight to the score.

| Test                          | Result                                          | Verdict |
|-------------------------------|-------------------------------------------------|---------|
| Bootstrap CIs on all 4 deltas | All exclude zero                                | ✓       |
| Window sweep (w=1 to w=6)     | Best concat recovers only 43.6% of gap          | ✓       |
| Prefixed-concat format        | Silhouette gap 0.090 > 0.05 threshold           | ✓       |
| Tz-position permutation       | Every concat ordering below mean-pool spoof AUC | ✓       |
| Trivial baseline comparison   | Mean-pool +0.119; concat +0.032 over trivial    | ✓       |

Score: 7/7 — confirmed across three independent investigations.

## ! Critical Finding: Training Configuration Matters

An initial investigation reached the **opposite conclusion** — concat beat mean-pool on every metric. Root cause: a degenerate training configuration causes silent within-feature embedding collapse. All timezone values converge to near-identical vectors (sim = 0.9993), eliminating the signal mean-pool depends on.

| Configuration                  | Within-feature sim | Spoof AUC                     | Conclusion          |
|--------------------------------|--------------------|-------------------------------|---------------------|
| CBOW + per-event corpus        | 0.9993 — collapse  | 0.384 ( <i>below chance</i> ) | Mean-pool refuted   |
| Skip-gram + per-account corpus | 0.392 — healthy    | 0.869                         | Mean-pool confirmed |

**This failure is silent.** Novel and fleet AUC look reasonable even under collapse — only a within-feature token similarity check (T8) detects it. Run after every retraining cycle; alert if similarity > 0.5.

## Recommended Production Architecture

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```
At login (<1ms):
tokens      = [os_X, br_X, tz_X, lang_X, net_X, sc_X]
embedding   = mean(fasttext[t] for t in tokens)
risk_score  = cosine_distance(embedding, account_centroid)
→ step-up auth if risk_score > threshold

Monthly retraining:
corpus: one sentence per account (all login events concatenated)
FastText: sg=1, epochs=20, negative=10, window=6, vector_size=64
✓ Health check: within-feature cosine sim – halt deployment if > 0.5

Offline (daily/weekly):
device_id_history → Word2Vec centroid → fleet/reuse review queue

Fallback (<20 events or service unavailable):
per-account known-device hash set → step-up if unseen profile
```

**Risk to monitor:** The mean-pool spoof advantage over the trivial baseline is +0.119 — operationally meaningful. Include spoof-specific AUC in any A/B evaluation before cutover.